Technische Hochschule Würzburg-Schweinfurt
Faculty of Computer Science and Business Information Systems

Master's Thesis

Text2Cypher: Autonomous Cypher Query Generation for Knowledge-Graph-Grounded Quality Process Improvement at BMW Group

Submitted to the Technische Hochschule Würzburg-Schweinfurt, Faculty of Computer Science and Business Information Systems, in partial fulfilment of the requirements for the degree of Master of Science in Artificial Intelligence

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Written in cooperation with BMW Group

[1]

Abstract

Write last. About 200–300 words: context, problem, approach, key results, conclusion.

Acknowledgements

TODO

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List of Acronyms

8D Eight Disciplines Problem Solving

AI Artificial Intelligence

API Application Programming Interface

EM Exact Match

FMEA Failure Mode and Effects Analysis

KG Knowledge Graph

LLM Large Language Model

NL Natural Language

QMT Quality Management (BMW Group)

RAG Retrieval-Augmented Generation

RCA Root Cause Analysis

RPN Risk Priority Number

SQL Structured Query Language

1 Introduction

Short opening: why quality engineers need to query 8D and FMEA knowledge, and why that is hard today.

1.1 Motivation

In the automotive industry, product quality is a decisive factor for customer satisfaction, brand reputation and long-term business success. Increasing product complexity, shorter innovation cycles and the ongoing transition towards electrification, software-defined vehicles and connected-car technology continue to increase the challenges associated with quality assurance. At the same time, customers expect premium-quality products and near-zero defect rates throughout the entire vehicle lifecycle.

To meet these expectations, automotive manufacturers rely on extensive quality management processes that completely span the supply chain, from the initial design and development phase to production, delivery and after-sales service. A significant portion of vehicle value is created by external suppliers, making effective supplier quality management essential. Quality issues originating from supplier components can lead to production disruptions, warranty claims, customer dissatisfaction and, in severe cases, vehicle recalls.

Within supplier quality management, the Eight Disciplines Problem Solving (8D) methodology represents one of the most important problem-solving approaches. Whenever a significant quality deviation is reported, suppliers are required to document the issue using an 8D report. These reports contain detailed information regarding problem descriptions, containment actions, root cause analysis, corrective measures and effectiveness checks. As a result, thousands of valuable problem-solving experiences and lessons learned are accumulated over time.

In parallel, Failure Mode and Effects Analysis (FMEA) serves as a preventive quality planning method. FMEAs are created to identify potential failure modes, evaluate associated risks and define preventive mitigation measures before failures occur. While 8D reports capture realized quality issues, FMEAs describe anticipated risks and preventive

controls. Together, both artifacts represent complementary sources of organized quality knowledge.

However, despite the availability of large amounts of quality-related data within modern enterprise systems, this knowledge is often underutilized and distributed across multiple databases, applications and document repositories. Information stored in 8D reports, FMEAs, supplier systems and other quality management platforms frequently exists in isolated data silos with limited semantic connections between them. As a result, identifying relationships between recurring defects, materials, suppliers, root causes and preventive risk assessments remains a largely manual and time-consuming process.

This fragmented data landscape significantly limits the ability to perform advanced analytics and exploit modern Artificial Intelligence (AI) technologies. Traditional relational data structures are effective at storing information, but often struggle to represent the complex interconnected relationships that exist within quality management processes.

Knowledge Graphs (KGs) provide a promising approach to addressing this challenge. By explicitly modelling entities such as suppliers, materials, defects, root causes, corrective actions and FMEA risks together with their relationships, KGs create a connected data foundation that reflects the real-world dependencies within supplier quality processes. Such a connected knowledge layer enables advanced querying, holistic risk analysis and traceability across previously disconnected datasets.

Furthermore, KGs establish the foundation for AI-driven quality management applications. By combining structured quality knowledge with Large Language Models (LLMs) and Agentic AI systems, it becomes possible to support engineers in tasks such as root cause investigation, identification of uncovered risks, supplier risk assessment and retrieval of relevant historical problem-solving knowledge.

Against this background, this thesis investigates how quality-related data from 8D reports, FMEAs and associated quality systems can be integrated into a KG-based architecture and how such a connected knowledge foundation can support intelligent analytics and AI-assisted decision-making within supplier quality management. With large volumes of quality data already existing within BMW's quality landscape, the lack of semantic connectivity between datasets limits their reuse for cross-domain analytics and AI applications. Establishing a KG-based quality knowledge foundation therefore represents a key step towards enabling intelligent supplier quality management and Agentic AI-driven decision support.

Example of acronyms: a LLM translates a question in Natural Language (NL) into a query on a KG. The second time, LLM and KG appear in short form only. Plural: LLMs.

Example of a figure reference: figure 1.1 shows ... Example of a citation [1].

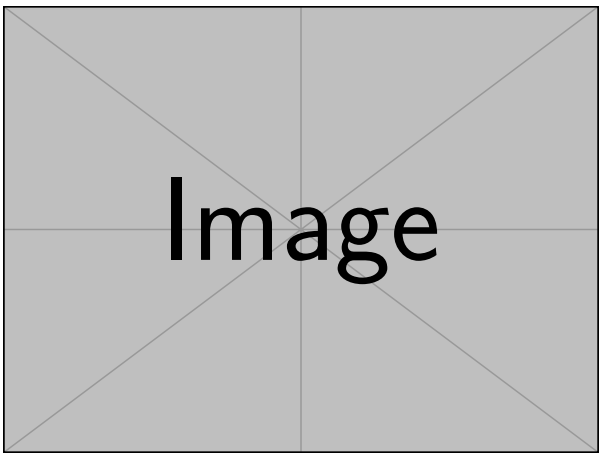


Figure 1.1: Placeholder figure.

1.2 Research Questions

RQ1 TODO

RQ2 TODO

RQ3 TODO

1.3 Contributions

TODO

1.4 Structure of this Thesis

Chapter 2 introduces ... Chapter 3 describes ... Chapter 4 presents ... Chapter 5 evaluates ... Chapter 6 concludes ...

2 Fundamentals

TODO: one paragraph saying what this chapter covers.

2.1 Quality Management with 8D and FMEA

2.1.1 The 8D Method

TODO

2.1.2 Failure Mode and Effects Analysis

TODO

2.2 Knowledge Graphs and Neo4j

TODO

2.3 Large Language Models and Text-to-Query

TODO

3 Problem Statement

TODO

3.1 Current Situation

TODO

3.2 Requirements

ID	Type	Requirement
R1	Functional	TODO
R2	Functional	TODO
R3	Non-functional	TODO

Table 3.1: Requirements for the system.

4 Solution

TODO

4.1 Graph Schema

Listing 4.1 shows an example Cypher query.

```
// Find failure modes linked to a component
MATCH (c:Component {name: $component})<-[:AFFECTS]-(f:FailureMode)
OPTIONAL MATCH (f)-[:CAUSED_BY]->(cause:Cause)
RETURN f.description AS failureMode,
       collect(DISTINCT cause.description) AS causes
ORDER BY failureMode
LIMIT 10
```

Listing 4.1: Example Cypher query (placeholder).

4.2 Agent Architecture

Listing 4.2 shows an example in Python.

```
def run_query(driver, cypher: str, params: dict) -> list[dict]:
    """Execute a Cypher query and return the records."""
    with driver.session() as session:
        result = session.run(cypher, params)
        return [record.data() for record in result]
```

Listing 4.2: Example Python code (placeholder).

A prompt can be shown with the `prompt` style:

```
You are an expert in Neo4j Cypher. Given the schema below,
translate the user's question into a single Cypher query.
```

Listing 4.3: Example prompt (placeholder).

5 Evaluation

TODO

5.1 Experimental Setup

TODO

5.2 Results

Table 5.1 shows how a results table can look.

Configuration	Accuracy	Valid queries	Latency (s)
Baseline	0.00	0.00	0.0
Variant A	0.00	0.00	0.0
Variant B	**0.00**	**0.00**	0.0

Table 5.1: Placeholder results table.

5.3 Discussion

TODO

6 Conclusion

TODO

6.1 Summary

TODO

6.2 Future Work

TODO

Appendix

Appendix A: Example Table

Heading 1	Heading 2
Title 1	Description 1
Title 2	Description 2

Table 1: Placeholder appendix table.

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Bibliography

[1] E. Gamma et al. *Entwurfsmuster: Elemente wiederverwendbarer objektorientierter Software*. Programmer's choice. Pearson Deutschland, 2011.

Statutory Declaration

I hereby declare that I have written this Master's thesis independently and have not submitted it elsewhere for examination purposes. All sources and aids used have been indicated, and all direct and indirect quotations have been marked as such.

Harsha Sathish, October 12, 2026

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